

22AIE111 – Object Oriented Programming in Java

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

Case Study – Phase I (Analysis)

Problem Identification & Class Mapping | CO1

Case Study No.: 10 / 32

Group No.: 10

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1. Problem Title

Vehicle Rental Management System

2. Problem Description

Manual vehicle rental processes result in booking overlaps, poor tracking of vehicle availability, and inefficient rental charge calculations. A digital system is required to manage vehicle bookings, automate charge computation, and maintain accurate records of vehicle usage — ensuring smooth operations for customers, staff, and administrators.

3. Identified Entities (Classes)

Class Name	Description
Vehicle	Represents a rentable vehicle — stores type, model, rate per day, and availability status.
Customer	A registered user who can search vehicles, place bookings, and view past rentals.
Booking	Records a rental transaction linking a Customer to a Vehicle for a given duration.
Staff	An employee responsible for managing vehicle records, updating status, and processing returns.
Admin	Oversees the entire system — manages users, monitors all bookings, and generates reports.

Class Name	Description
RentalCharge	Handles calculation of rental cost based on duration and vehicle rate; produces invoices.
Inventory	Maintains the list of all vehicles and tracks real-time availability across the fleet.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods / Behaviours
Vehicle	vehicleId model type ratePerDay isAvailable	getDetails() setAvailability() getRate()
Customer	customerId name contactInfo rentalHistory	register() searchVehicle() makeBooking() viewHistory()
Booking	bookingId startDate endDate customer vehicle status	confirmBooking() cancelBooking() getDuration()
Staff	staffId name role	addVehicle() updateVehicle() processReturn()
Admin	adminId name accessLevel	manageUsers() generateReport() viewAllBookings()
RentalCharge	chargeId booking totalAmount paymentStatus	calculateCharge() applyDiscount() generateInvoice()
Inventory	vehicleList totalCount availableCount	checkAvailability() updateInventory() getAvailableVehicles()